# Achyuthan Sivasankar

achyuthan.sivasankar@gmail.com | linkedin.com/in/achyuthan-s | github.com/achyuthan-s | achyuthan.dev |
New York, NY

ML Research Engineer · Adaptive Routing & Sparse Architectures · MoE / KAN Systems · NYU MS CS (2025–2027) · 3 IEEE publications · 1 Patent

## Education

**New York University — Courant / Tandon** New York, NY
*M.S. Computer Science* — GPA: 4.0 | Algorithms (A), Deep Learning (A), HCI (A) Aug 2025 – May 2027

- Research assistant, **Prof. Anna Choromanska's lab** (Summer 2026) — investigating loss landscape geometry and convergence behaviour of sparse, routing-based neural architectures; connects directly to MoE expert-collapse and KAN routing work.
- Relevant coursework: Algorithms (A), Engineering Deep Learning (A), Artificial Intelligence, HCI (A).

**Vellore Institute of Technology** Vellore, India
*B.Tech Computer Science* — GPA: 3.87 / 4.0 Aug 2021 – May 2025

- Three IEEE publications (2024): video steganography, acoustic event detection, retail ML forecasting.
- Patent filed (VIT IPRTT Cell, 2025) — ML-enhanced blackhole-attack detection in IoT RPL networks.

## Technical Skills

| | |
|---|---|
| **ML / Research** | PyTorch, JAX, HuggingFace, MoE architectures, KAN layers, LoRA/PEFT, RAG, LLM fine-tuning, RL (REINFORCE, Meta-World), Computer Vision, NLP, Self-supervised learning |
| **Languages** | Python (expert), C/C++, Go, SQL, Bash |
| **MLOps & Infra** | Docker, FastAPI, Flask, Redis, PostgreSQL, Neo4j (ChromaDB), AWS (S3, EC2), Weights & Biases, Git, Linux |
| **Frameworks** | TensorFlow / Keras, Scikit-learn, Pandas, NumPy, Streamlit, Optuna |

## Research

**KAN-Multi: Adaptive Multi-Basis Routing with Emergent Specialization** | First Author — Manuscript in preparation for workshop submission

- *Core thread:* Part of a unified research program on **adaptive routing in sparse architectures** — KAN-Multi (function-basis routing), MoE-Bench (expert routing diagnostics), and AutoMoE (topology routing via meta-learning) address the same fundamental question at different scales.
- Designed a statistics-driven routing layer selecting among **6 function bases** (Fourier, Wavelet, Chebyshev, RBF, Rational, Sigmoid) with **zero routing supervision**; emergent specialization measured via per-layer Shannon entropy dynamics across training.
- **+6.8% over MLP baseline on CIFAR-100** (MLP: 19.0%; KAN-Multi: 25.8%) and **+1.2% on Adult tabular** ($p$<0.001, 5-seed, matched parameter budgets); task-complexity scaling law $c\sqrt{k(f)}$ predicts KAN advantage ($r$=0.972, $p$=0.006).
- Hybrid ResNet-18 + KAN head: **92.42% on CIFAR-10** (vs. 91.8% linear baseline), **92.51% on Fashion-MNIST**; negative result on ResNet-50/CIFAR-100 characterised and included — identifies regime boundaries for adaptive routing, not suppressed.
- Soft→hard routing: **theoretical 5–6× FLOP reduction** at low router entropy; PyTorch wall-clock bottlenecks documented rigorously; CUDA/Triton path identified as future work.

**MoE-Bench: Open Benchmark for Expert Collapse & Routing Efficiency in Sparse MoE LLMs** | First Author — Open-source toolkit + paper

- Built **pip-installable evaluation toolkit** for routing entropy, expert utilisation, and collapse across OLMoE-1B-7B, JetMoE-8B, and Qwen1.5-MoE-A2.7B with **domain-matched prompts, bootstrap CIs, and a Gradio demo**.
- Identified **50% layer-collapse** on OLMoE math domain ($\tau$=0.40) vs. 0% on JetMoE/Qwen — demonstrating collapse is **architecture-dependent**, not universal.
- Implemented $L_{div}$ LoRA training on OLMoE with MMLU evaluation; contributed novel entropy-floor regulariser: $\max(0, \tau - H_l)$ per collapsed layer.

**AutoMoE: Meta-Learning Expert Topologies for Multi-Task Robotic Manipulation** | Research Project

- Evolutionary REINFORCE meta-learner over discrete MoE topology space (1–3 tiers × 2–5 experts/tier); eliminates hand-designed architectures.

- Discovered topology **(3,3,2) achieves 100% task success at 11.5M params** vs. 23.3M for fixed Hierarchical MoE — **2× parameter efficiency** at equal performance.
- Validated on **Meta-World ML10** (10 robotic manipulation tasks); warm-start convergence within 20 inner epochs per topology candidate.

## Experience

**Indian Institute of Science (IISc)** | Research Intern — Bangalore, India — Jun 2024 – Aug 2024

- Applied **GNN-based graph-theoretic optimisation** to large-scale RF mesh networks (1k+ nodes); routing efficiency improved **42%** over shortest-path baseline and intrusion detection improved **28%** over rule-based IDS on the same network topology set.
- Improved EEG signal classification for neurodegenerative disease diagnosis by **18%** over prior CNN baseline on a 500k-signal BCI dataset, using attention-augmented deep learning.

**National University of Singapore (NUS)** | AI Research Intern — Singapore — Dec 2023 – Feb 2024

- Built real-time multimodal stress-detection system achieving **99.67% accuracy on SWELL-KW** (prior SOTA: ∼97%; unimodal baseline: ∼94%) with multi-threaded architecture reducing inference latency by **35%** vs. single-threaded implementation.
- Prototyped VR-driven therapeutic platform integrating generative AI, real-time emotion recognition, and dynamic music recommendation (Spotify API, 126k+ tracks, KNN-based personalisation).

*Earlier: ML Security Engineering Intern, Cyber's Secret, Abu Dhabi (Aug–Oct 2023) — ML-augmented log-analysis pipeline, 50k+ logs/sec, 27% accuracy gain over rule-based baseline; directly motivated subsequent IoT IDS and patent work.*

## Selected Projects

**ACGA: Adaptive Corrective Graph-Augmented RAG System** | *Python, FastAPI, Redis, ChromaDB, Neo4j, PostgreSQL, Docker*

- Applies **adaptive routing principles** from MoE research to retrieval: a query router selects dynamically among vector, graph, and keyword retrieval strategies per query type — the same routing-efficiency problem studied at the model level in KAN-Multi and MoE-Bench.
- **4-layer memory architecture** (Working / Session / Episodic / Semantic) addresses long-term context retention — the critical open gap in production RAG systems.
- **85ms cold-query latency**, **15.5× cached speedup** (200 QPS); MIT-licensed, Dockerised, production-ready; 0-cost deployment on Gemini free tier.

**IoT Blackhole-Attack IDS** | *Python, LSTM, Random Forest, XGBoost, Contiki-NG/COOJA*

- LSTM + ensemble classifier achieving **98% detection accuracy, 96% precision** across 30+ network topologies in Contiki-NG simulation; false-alarm rate **reduced to 3%**.
- Underlying ML system awarded **Indian Patent** (VIT IPRTT Cell, 2025).

## Patents & Publications

- **Patent** — "ML-Enhanced Intrusion Detection for Blackhole Attacks in IoT RPL Networks," VIT IPRTT Cell, India, 2025.
- **IEEE (×3, 2024)** — Video steganography with AES-256/RSA-4096; CNN-based acoustic event detection (MFCC); retail ML demand forecasting.
- **Manuscript in preparation** — "KAN-Multi: Adaptive Multi-Basis Routing with Emergent Specialization" (first author); targeting workshop venue, 2026.
- **Open toolkit** — MoE-Bench: routing entropy & expert-collapse benchmark for sparse MoE LLMs (OLMoE, JetMoE, Qwen); Apache-2.0; github.com/moe-bench/moe-bench.